Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Core languages used to process, transform, calculate features, and programmatically serve analytics endpoints.	Python 3.14, HTML5, CSS3, JavaScript (ES6)	Python enables efficient vectorization and data manipulation via Pandas; HTML/CSS/JS construct a clean responsive executive web shell.
2.	Database / Data Warehouse	Structured file-based data store containing anonymized patient clinical records and lifestyle metrics.	Flat CSV (smart_healthcare_dataset.csv), Pandas DataFrames	Provides portable, lightweight, high-performance data loading for 5,000 patient records without external database overhead.

3.	Data Ingestion	Methods and connections used to extract and ingest raw clinical vitals into analytical engines.	Python File I/O, Tableau Native CSV Data Connector	Ensures smooth data ingestion into Python Pandas for cleaning and directly into Tableau Desktop for hyper-extract building.
4.	Data Transformation and Business Logic	Scripting logic and calculated fields used to stratify demographics, engineer risk indices, and aggregate KPIs.	Pandas Feature Engineering, Tableau Calculated Fields	Calculates Age Brackets (18–34 to 65+), WHO BMI Categories, and the multi-variable Dynamic Risk Tier Index (High/Moderate/Low).
5.	Data Visualization / Analytics	Business Intelligence engine used to design interactive worksheets, dashboards, and executive stories.	Tableau Desktop / Tableau Public	Renders 9 visual worksheets, 2 master dashboards, and a 4-scene guided Executive Story for deep clinical risk evaluation.
6.	Front-End / User Interface	Web hosting framework and embedding API layer that serves the interactive dashboard to end-users.	Python Flask Framework (app.py), Tableau JavaScript API v2	Serves a lightweight web application locally on http://127.0.0.1:5001 with embedded interactive Tableau views and real-time KPI stat cards.

Figure 1: Health360 Technology Stack & Deployment Architecture

